Fall 2025 – ECE 302: Probabilistic Methods in ECE

Course Information

- **Course number and title:** 20124 - ECE 30200 - 005: Probabilistic Methods in ECE
- **Instructor:** Prof. Abolfazl Hashemi

- **Instructional Modality:** in person
- **Course credit hours:** 3 credits
- **Prerequisites:** (Undergraduate level MA 26200 Minimum Grade of D- or Undergraduate level MA 26600 Minimum Grade of D- or Undergraduate level MA 36600 Minimum Grade of D- or Undergraduate level MATH 26600 Minimum Grade of D-) and Undergraduate level ECE 30100 Minimum Grade of D- [may be taken concurrently]

- **The course website on Brightspace: https://purdue.brightspace.com/d2l/login**

Course Description

This course is intended to introduce the concepts of probability and random processes. The tools of probability theory, and of the related field of statistical inference, are the keys for being able to analyze and make sense of data. These tools underline important advances in many fields, from the basic sciences to engineering and management.

Learning Resources, Technology & Texts

- **Main Resource**
  - Slides, annotated slides, and lecture notes provided by the instructor based on the main textbook below
  - Please see the detailed schedule on the last page.

- **Main Textbook**
  - Bertsekas, Dimitri, and John Tsitsiklis. *Introduction to Probability.* 2nd ed. Athena Scientific, 2008. ISBN: 9781886529236.
- **Additional References**
  - Probability, Statistics, and Random Processes for Electrical Engineering, 3rd Edition, Alberto Leon-Garcia, Prentice-Hall, 2008, ISBN No. 9780131471221
  - Introduction to Probability for Data Science, Stanley H. Chan, Michigan Publishing, 2021, ISBN 978-1-60785-746-4

- **Use of artificial intelligence (AI) or Large Language Models (LLM) in this course**
  - Students may use AI and LLM to better learn the materials. For instance, you may upload the lecture notes and slides on your favorite LLM and ask it to explain topics you are struggling with or provide additional examples.
  - You may not upload HWs and their solutions, however.
  - Any submitted assignments should be the work of the student alone and LLMs are not allowed.

- **Software/Websites**
    - This term we will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates and the instructors. The class on piazza is structured so you can discuss each homework and topics on each exam. Please post to the relevant thread to ensure a proper response.
    - To honor and respect one another, it is important to adhere to the netiquette posted in the Start Here section of the Brightspace course.

        - You will receive an invitation to join piazza. If you register late for the course or if you don't receive the email, please go to piazza.com and register for the class yourself using the link.
- **Academic Success and/or Tutoring support**
    - Information for Technology Resources, Academic Help and Campus Resources can be found under Student Support and Resources in the Table of Contents in the Brightspace course
- **Brightspace learning management system (LMS)**
    - Your ECE302 Probabilistic Methods in ECE is housed in the Brightspace LMS

Course Learning Outcomes

A student who successfully fulfills the course requirements will have demonstrated:

- an ability to solve elementary probability problems involving random events and random variables.
- an ability to model uncertainty by random variables and analyze the implications in a range of engineering applications.
- an understanding of the idea of a random process, along with some basic examples and applications.

Grading Policy

- Homework: 35% **(8 sets, each 5%; the HW with the lowest score will be dropped)**
- Midterm 1: 15% **(Scheduled for October 2, in class)**
- Midterm 2: 15% **(Scheduled for November 13, in class)**
- Final Exam: 35% **(Scheduled for TBD)**

Grading Scale

Your final class letter grade will be determined based on the following approximate cut-offs (range includes minus and plus grades as well). Depending on the average class performance, these cutoffs may be modified up to, typically, 5% in either direction, or curving mechanisms may be used.

Grade Ranges	Percentage
A Range	90-100
B Range	80-89.99
C Range	65-79.99
D Range	55-64.99
F Range	0-54.99

Absences

Y should contact the TAs if something arises in your personal life that will impact your ability to complete your coursework. University regulations concerning absences are updated on the Office of the Dean of Students website.

If you have a death in the family, you (or your representative) should contact the Office of the Dean of Students. That office will then notify your instructors. Check the Grief Absence Policy on Purdue's website.

If you are called up for active duty or mandatory military training, your commanding officer can provide the Dean of Students a copy of your orders. The Dean of Students may then contact your instructors; however, please let Prof. Hashemi know at the beginning of the semester if you expect to be absent for military duty.

If you need to adjust deadlines for a religious observance, you must let Prof. Hashemi know a week in advance, minimize the length of your absence, and be flexible in arranging alternative times to complete the assignments you may miss.

For all absences related to COVID or other infectious diseases, please see the Protect Purdue website.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern.

The Student Widget includes a link to Academic Policies & Regulations, which links to Student Conduct and includes a Statement of Integrity and Code of Conduct.

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

See Purdue's full Nondiscrimination Policy Statement.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone: 765-494-1247, as soon as possible.

If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: **https://www.purdue.edu/drc/students/course-accommodation-letter.php** Additionally, you are strongly encouraged to contact me as soon as possible to discuss implementation of your accommodations.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a new web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to students, faculty, and staff at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students (ODOS). Call 765-494-1747.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the session. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's

control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed, and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday.

Disclaimer

The instructor reserves the right to update or change the syllabus when and where needed. You will be notified by email or announcements of any changes made.